Experiment 7

ELECTRON SPIN RESONANCE BY OPTICAL PUMPING

(a) Theory. ,

The following material is reproduced from Experiment 6, Electron Spin Resonance (ESR) by Microwave Absorption, since it is relevant here also:

Quantum mechanically an electron spin can take on only certain directions (energies) in the presence of an external magnetic field. If a difference in the number of electrons in two energy states exists, it is possible to determine the energy separation by a resonance technique. The two common techniques for observing ESR are microwave absorption and optical pumping.

(a) ESR by Microwave Absorption Utilizing the Maxwell-Boltsmann Population Difference. A very small difference between the populations of two closely-spaced energy levels in thermal equilibrium always exists and is given by the Maxwell-Boltzmann distribution. Often the temperature is lowered or the field increased to enhance this population difference. If microwave radiation of the correct frequency is applied to the sample, microwave power is absorbed tending to equalize the populations.

(b) ESR by Optical Pumping and Detection. In this case the technique of optical pumping produces a population difference many times (100 or more) larger than the Maxwell-Boltzmann difference. Again rf or microwave power is absorbed but the decrease in population difference is observed by noting the change in optical absorption.

(c) Comparison of Microwave and Optical ESR. Since optical frequencies (about 10^{14} cps) are much higher than the usual microwave resonance frequencies (about 10^{10} cps), an energy amplification of about 10^4 results. Combined with the factor of 100 from optical pumping, the optical method is about 10^6 times more sensitive than the microwave method. Whereas the microwave absorption method is usually limited to samples with 10^{15} or 10^{16} atoms/cm^3 or more, the optical method works well for as few as 10^9 or 10^{10} atoms/cm^3. However, the optically detected ESR samples must usually be in the form of a gas of density less than 10^{14} atoms/cm^3 in order to avoid complete absorption of the light used to detect the resonance. Also resonance radiation must bc available and the sample pumpable. This greatly limits the candidates for optical ESR. On the other hand, the microwave absorption technique is applicable to a wide variety of samples. Sometimes a hybrid of optical pumping and microwave detection is found useful.

(d) Applications of ESR. Microwave ESR has been utilized in studying many substances with unpaired electrons, in particular the free radicals important in the chemical activities of living cells. In fact, ESR has been observed in living cells.

Optical ESR has been used to construct very precise frequency standards and magnetometers and to study various properties of paramagnetic gases..

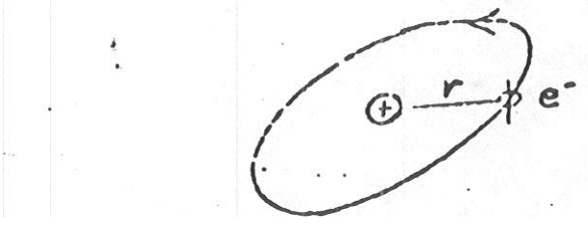
ESR BY MICROWAVE ABSORPTION

(a) General Theory

The following description of ESR-was prepared by Richard Ballard.

"It is possible to picture an atom semiclassically as a positively charged and massive nucleus surrounded by negative electrons orbiting under the influence of an electrostatic attraction. Experiment and quantum theory have placed limitations and added new elements to this simple picture. It is known for instance that only certain electron orbits are allowed. and that aside from this orbital motion, the electrons possess an intrinsic spin.

Schematically, a simple atom such as hydrogen might be drawn as follows:



The angular momentum of the electron in its orbit is given by

$$L = mvr = m\omega r^2 = m\frac{2\pi}{T}r^2$$

where T is the period of the orbit.

Electrically the orbiting charged electron gives the atom the appearance of having a current loop with a charge -e/c (esu units) passing by each point every period, i.e.. $i = -(e/c)/T$. This current loop gives rise to a magnetic moment μ_L, where

$$\mu_L = iA = \frac{-e}{cT}\pi r^2 = -\left(\frac{e}{2mc}\right)\frac{m2\pi r^2}{T} = \frac{-e}{2mc}L \ .$$

$$\mu_L = +\gamma L \quad \text{where } \gamma \equiv \frac{-e}{2mc} \text{ (gyromagnetic ratio) . ,}$$

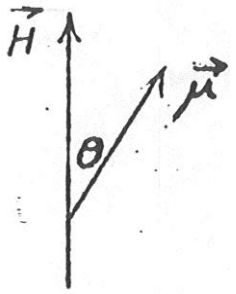
The gyromagnetic ratio relates the magnetic moment of the orbital motion to the orbital angular momentum. In much the same way, it is possible to write down the total magnetic moment μ in terms of the total angular momentum J, (units $\hbar$), including the contributions of both the spin and the orbital motion,

$$\mu = \gamma_J J = g_J \beta J/\hbar \ , \quad \beta = q\hbar/2me \ ,$$

m = mass of the particle
q = charge of the particle ($\pm e$)

The so-called "g-factor" is introduced to account for certain precessional and relativistic effects; under varying- circumstances it may range from 0 to 15; $g_L = 1$ for orbital angular momentum; $g_S \approx 2$ fcr the spin of a free electron.

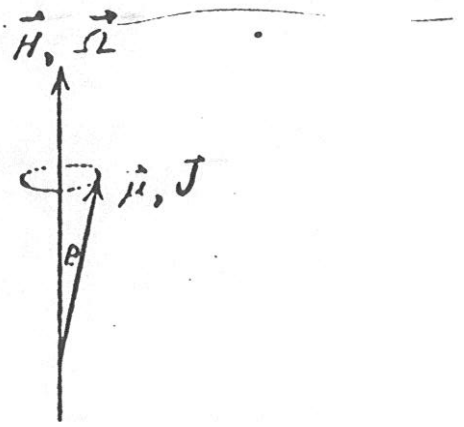
If an external magnetic field is applied to the atom, a torque τ is exerted on the magnetic moment tending to line it up with the field.



$$\vec{\tau} = \vec{\mu} \times \vec{H}$$

$$\tau = \mu H \sin\theta$$

The orbiting and spinning mass of the electron causes it to behave like a gyroscope under this torque, with the result that the magnetic moment precesses 'about the applied field with a precessional frequency Ω. As with a gyroscope this is given by



$$\vec{\tau} = \vec{\Omega}_L \times \vec{J}$$

$$\mu H \sin\theta = \Omega_L \sin\theta \, J$$

$$\Omega_L = \frac{\mu H}{J} = g_J \beta J H/J\hbar$$

$$\Omega_L = g_J \beta H/\hbar \text{ (Larmor angular frequency)}$$

$\frac{\Omega_L}{2\pi}$ Larmor frequency

Defining the zero of energy for $\theta = \frac{\pi}{2}$, the energy for an arbitrary angle θ is found by integrating the work done in moving μ against the torque τ.

$$E = \int_{\frac{\pi}{2}}^{\theta} \tau \, d\theta = \int_{\frac{\pi}{2}}^{\theta} \mu H \sin\theta \, d\theta$$

$$= -\mu H \cos\theta = -\vec{\mu} \cdot \vec{H} .$$

The quantum theory imposes certain restrictions upon this classical picture. First the total angular momentum J is given by:

$$|\vec{J}| = \sqrt{j(j+1)}\,\hbar$$

where the total angular momentum quantum number is restricted to

$$J = j\hbar \qquad j = \text{angular momentum quantum number (integer or half-integer)}$$

$$\hbar = h \text{ (Planck's constant)}/2\pi$$

Furthermore, the angles which $\vec{\mu}$ can make with respect to $\vec{H}$ are restricted so that

$$\cos\theta = \frac{m_j}{\sqrt{j(j+1)}} \quad \text{where } m_j = j,\ j-1,\ \cdots,\ -j .$$

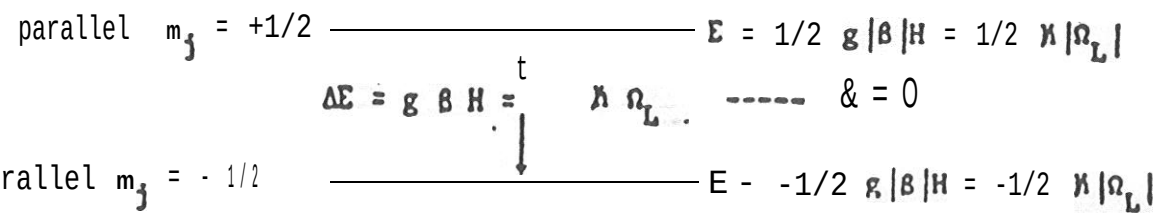
Consequently

$$E = -\mu H \cos\theta = -\mu H \frac{m_j}{\sqrt{j(j+1)}} = -\gamma_J \hbar H m_j \sqrt{j(j+1)} \,/\, \sqrt{j(j+1)}$$

$$E = -\gamma_J \hbar H m_j \qquad m_j = j,\ j-1,\ \cdots,\ -j \text{ (magnetic quantum number).}$$

$$E = -g_J \beta H m_j = -\hbar \Omega_L m_j, \text{ where } \beta = q\hbar/2mc \quad \text{(Bohr magneton)} .$$

As a result of the angular quantization, the electron spin-orbit system can only have energies which are multiples of $g\beta H = \hbar\Omega_L$. If, as in many cases, the atom has an angular momentum quantum number of $j = 1/2$, then m_j can only take on the values 1/2 or -1/2. In other words, the magnetic moment can be only parallel or anti-parallel. The energy diagram for $j = 1/2$ is given by



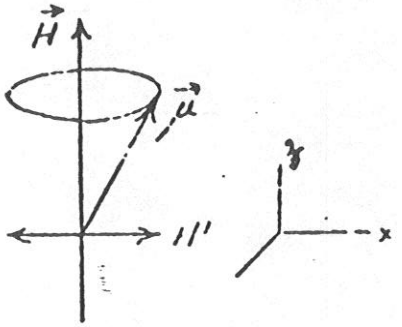
assuming B < 0 (q =

Statistical mechanics gives us further information about a collection of atoms with a thermal equilibrium temperature T. The ratio of the number of atoms (n_+) with $m_j = +1/2$ in the higher energy orientation (parallel) to the number (n_-) with the lower energy orientation is given by

$$\frac{n_+}{n} = e^{\frac{-\Delta E}{kT}} = e^{\frac{g\beta H}{kT}} = e^{-c|\beta|H/kT} \quad \text{for } \beta < 0.$$

Electron spin resonance is a technique by which, among other things, the energy splitting $\Delta E = g\beta H = \hbar\Omega$ can be measured. With a known field H the g-factor for various atoms can be found; similarly for an atom with known g-factor very accurate field measurements can be made.

Suppose that an alternating magnetic field $H' = H_o \cos \Omega t\, \hat{x}$ is applied perpendicular to the steady H field in which the atoms are precessing.



It is easy to show that any alternating field can be decomposed into two counter-rotating fields:

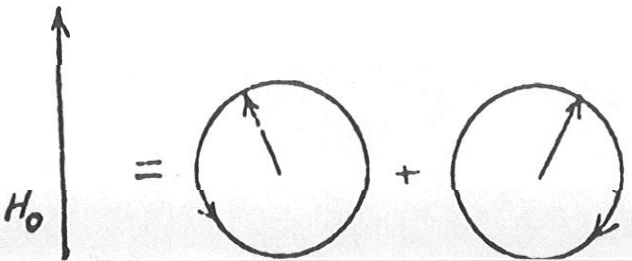
$$H_o \cos \Omega t\, \hat{x} = \frac{1}{2} H_o \cos \Omega t\, \hat{x} + \frac{1}{2} H_o \sin \Omega t\, \hat{y} +$$

$$\frac{1}{2} H_o \cos \Omega t\, \hat{x} - \frac{1}{2} H_o \sin \Omega t\, \hat{y}$$

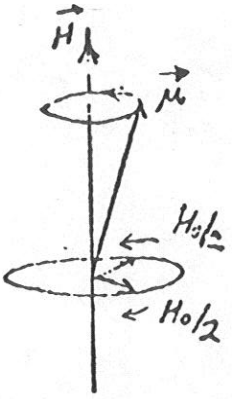
$$= \frac{1}{2} H_o (\hat{x} \cos \Omega t + \hat{y} \sin \Omega t)$$

$$+ \frac{1}{2} H_o [\hat{x} \cos(-\Omega t) + \hat{y} \sin(-\Omega t)]$$

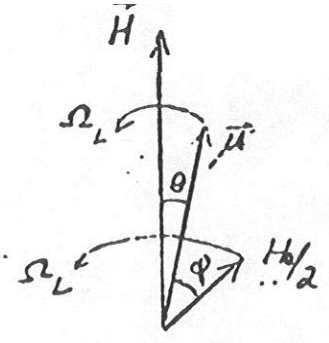
Pictorially:



Decomposing H' into two rotating fields at the atom, it is obvious that one field rotates in the same sense as the precessing magnetic moment while the other one rotates in the opposite sense.

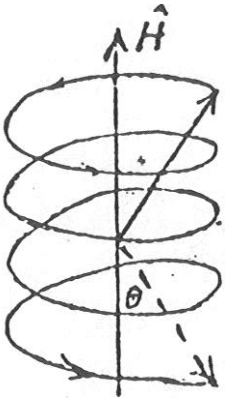


If the frequency of the imposed fields Ω is not equal to the precessional frequency of the atom Ω_L, then the position of the rotating fields will always be changing with respect to the magnetic moment. Consequently, the magnetic moment will only wobble slightly about its precessional path. If, however, the frequencies an? equal $\Omega = \Omega_L$, the rotation of one component of the field will just match the precession of the magnetic moment and the angle ϕ between



them will be constant. The rotating field will exert a torque $\frac{1}{2}\mu H_0 \sin\phi$ which will result classically in the magnetic moment also precessing about the rotating field.

If Ho << H, then the combined precessions will result in the magnetic moment



Period $T = 2\pi/(\gamma \frac{1}{2} H_o) = \frac{4\pi}{\gamma H_o}$

rotating rapidly at the Larmor frequency about H with the angle θ oscillating slowly through $\pi = 2\theta$ with a period T. -Since the energy of the spin system depends upon the angle θ, the-effect of the rotating field Ho is to cause some magnetic moments to gain energy by precessing to large θ angles and others to lose energy by precessing to smaller ones. Since quantum mechanics restricts the possible angles θ, the smooth classical oscillation will consist, rather, of discontinuous jumps exchanging a quanta of energy ($g\beta H = \hbar\Omega_L$) with the applied field.

The consequences of these jumps or transitions induced by the applied alternating field can be seen most easily in a two level ($J = \frac{1}{2}$) atom. If a material composed of such atoms is placed in crossed DC and AC magnetic fields, the number of magnetic moments aligned parallel or anti-parallel will be given by the Boltzmann factor

$$\frac{n_+}{n_-} = e^{+\frac{g\beta H}{kT}}$$

unless the AC frequency is adjusted to match the atom's Larmor frequency $\Omega_L = \gamma H$. As we have seen. when this happens magnetic moments will be induced to jump from parallel to anti-parallel orientations and vice versa. The probability w for induced jumps in one direction is the same for that in the other direction. Consequently the number of parallel moments going from parallel to anti-parallel is the number in that‘atate times the probability of a transition, $n_- w$. The number going in the other direction will be $n_+ w$. The net number going from low energy to high energy orientations and

absorbing energy is

$$n_- v - n_+ v = w(n_- - n_+) .$$

Since the total number N is constant,

$$N = n_+ + n_- = n_-(1 + \frac{n_+}{n_-}) = n_- (1 + e^{+\frac{g\beta H}{kT}})$$

$$n_- - n_+ = -N + 2n_- = N \left(\frac{2}{1 + e^{+\frac{g\beta H}{kT}}} - 1 \right)$$

$$n_- - n_+ = N \left(\frac{1 - e^{+\frac{g\beta H}{kT}}}{1 + e^{+\frac{g\beta H}{kT}}} \right) .$$

Since this formula predicts that there are more atoms in the lover energy level than in the upper, there will always be a net absorption of energy from the field.

To make $n_- - n_+$ large enough at room temperature it is necessary to use large fields of several kilogauss. Since the gyromagnetic ratio for most electron orbits is about 2 megacycles/gauss, the Larmor frequencies will be on the order of kilomegacycles (approximately X band microwaves). It is convenient-then to use the X band klystron and diode detector used in the 4C laboratory.

This completes the material reproduced from Experiment 6.

There are several excellent and relatively simple reviews of optical pumping: the articles by Bloom[1] and de Zafra[2] are probably adequate. Carver's[3] article on optical pumping and Bloom's[4] description of the Rb magnetometer are highly recommended also, particularly with regard to the applications of the technique. Reference 5 gives an extensive and complex treatment of optical pumping (much beyond the level expected here).

First you should understand optical pumping for a Rb atom with no nuclear spin. The energy level diagram of the ground state and first two

excited states of Rb is shown in Fig. 1. All of the levels shown have principal quantum number n = 5, i.e.. they all belong to the same Bohr orbit. An electron in a P state (with orbital angular momentum L = 1) is, on the average, farther from the rest of the atom than an electron in an S state (L = 0). The electrostatic energy is then higher ($-e^2/r$ is less negative) and the P state lies above the S state. The P state is also split by the spin-orbit interaction of the form $\vec{L} \cdot \vec{S}$. This splitting arises because the electron's magnetic moment can line up in two ways with respect to the field which the remainder of the atom produces at the electron (the electron "sees" the rest of the atom revolving around it). The transitions from the $^2P_{1/2}$ and $^2P_{3/2}$ states to the $^2S_{1/2}$ state are optical transitions at 7947 and 7800 Å, respectively.

If the Rb atom is subjected to an external magnetic field, $\vec{H}$, the electron's moment $\vec{\mu}$ interacts with the field causing a further splitting of the levels. The interaction energy is $-\vec{\mu} \cdot \vec{H}$. Quantum mechanics permits only certain orientations of $\vec{\mu}$ with respect to $\vec{H}$: only those for which the component along $\vec{H}$, M_J, of the total angular momentum $\vec{J}$ is an integer (if J is an integer) or a half-integer (if J is a half-integer). The difference between adjacent levels (Zeeman levels) in a magnetic field of 1 gauss is 2.0 Mc/sec. In actual fact, Fig. 1 is oversimplified. Rb^{87} has a nuclear spin I = 3/2; the nuclear and electronic moments interact through the hyperfine interaction $a\vec{I} \cdot \vec{J}$ to split the S and P states even in the absence of the external field (see the appendix). The difference between the Zeeman levels is then given by

$$\frac{\Delta\nu}{H} \approx \frac{2.8}{21 + 1} \frac{\text{Mc/sec}}{\text{gauss}} . \qquad (1)$$

Equation (1) is used to relate the magnetic field, the measured frequency and the nuclear spin. The optical pumping process can be understood using the simplified energy level diagram of Fig. 1. A narrow-band interference filter eliminates the 7800 Å D_2 line from the Rb lamp (the filter costs about $100, so be careful). The 7947 Å light from the lamp passes through a circular polarizer, filter, a lens, a bulb of Rb gas and buffer gas which is the absorption cell., and another lens and then impinges on a photocell, as shown in Fig. 2. The circularly polarized

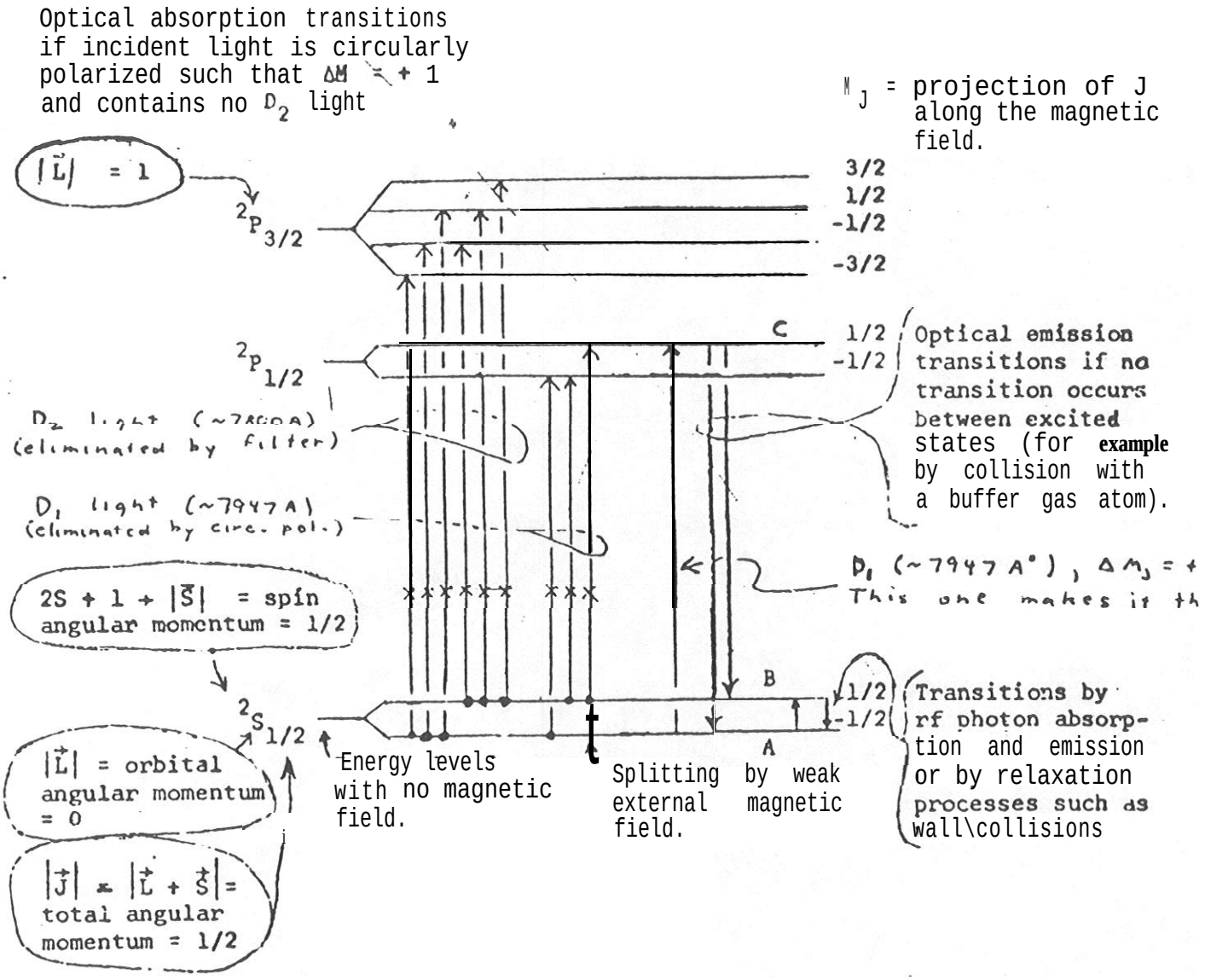
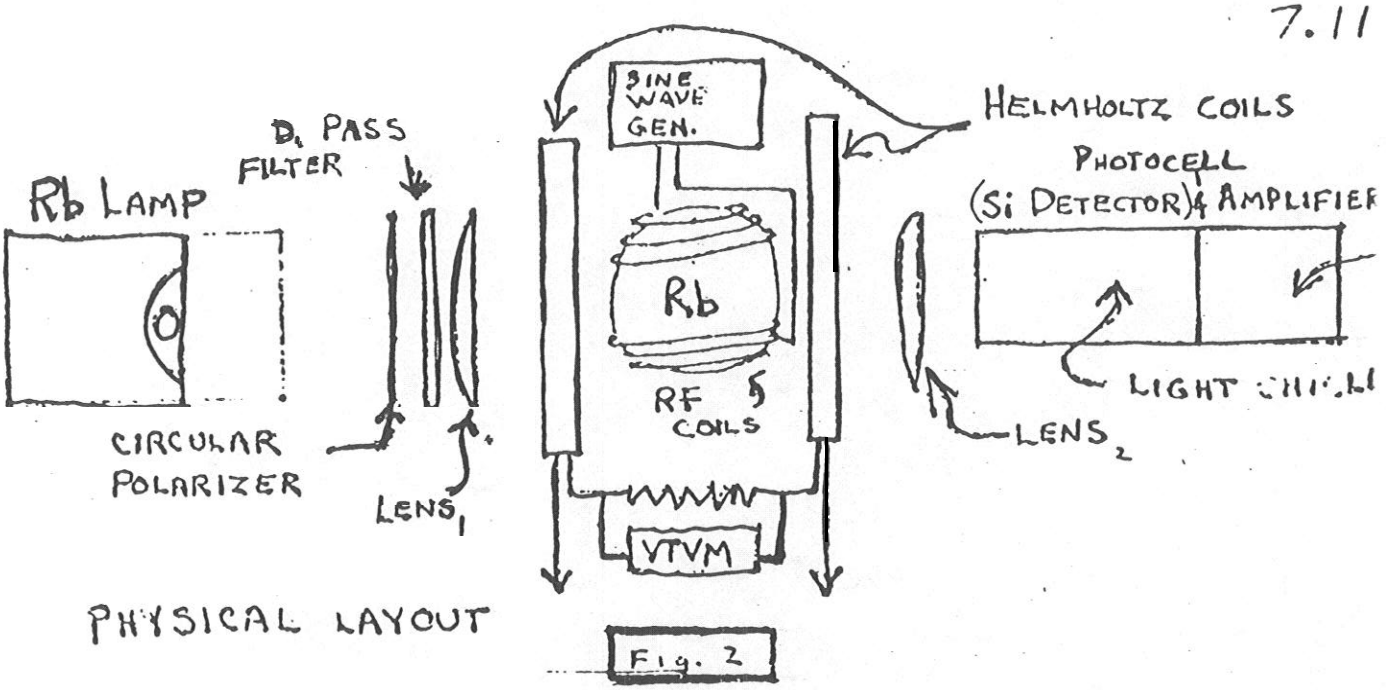


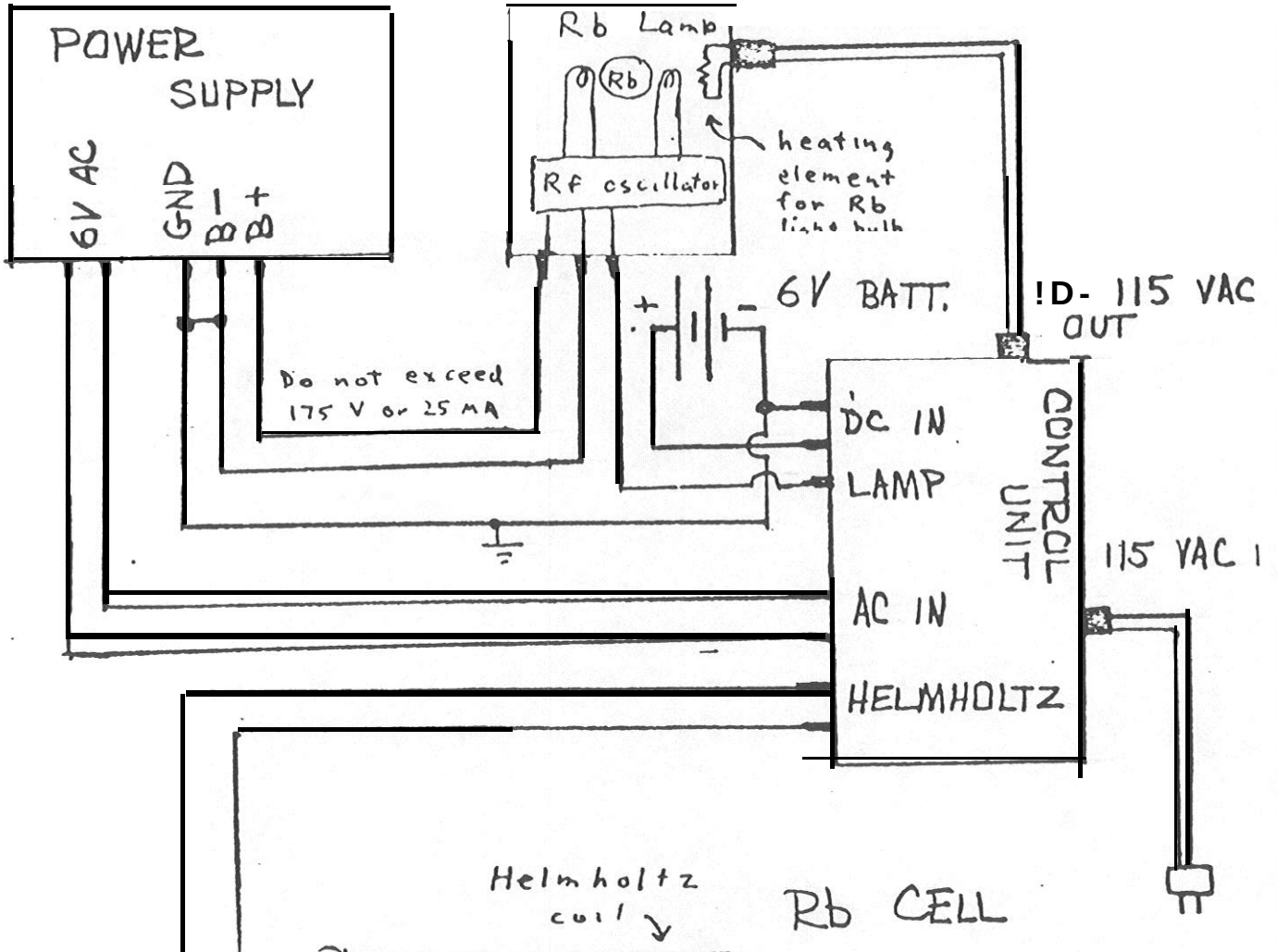
Figure 1

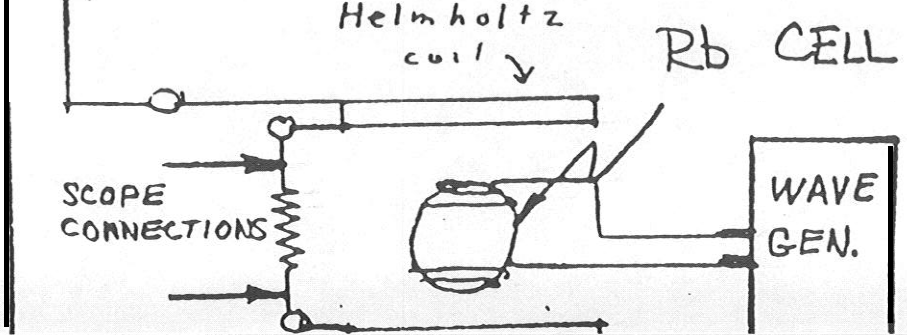
Energy Levels of Rubidium (Neglecting the Nuclear Spin) and the Optical Pumping Cycle



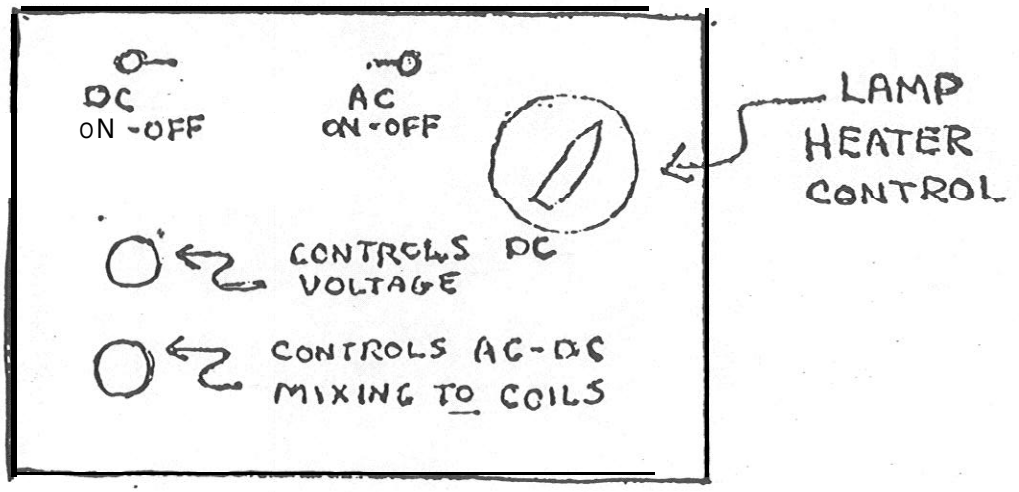
PHYSICAL LAYOUT

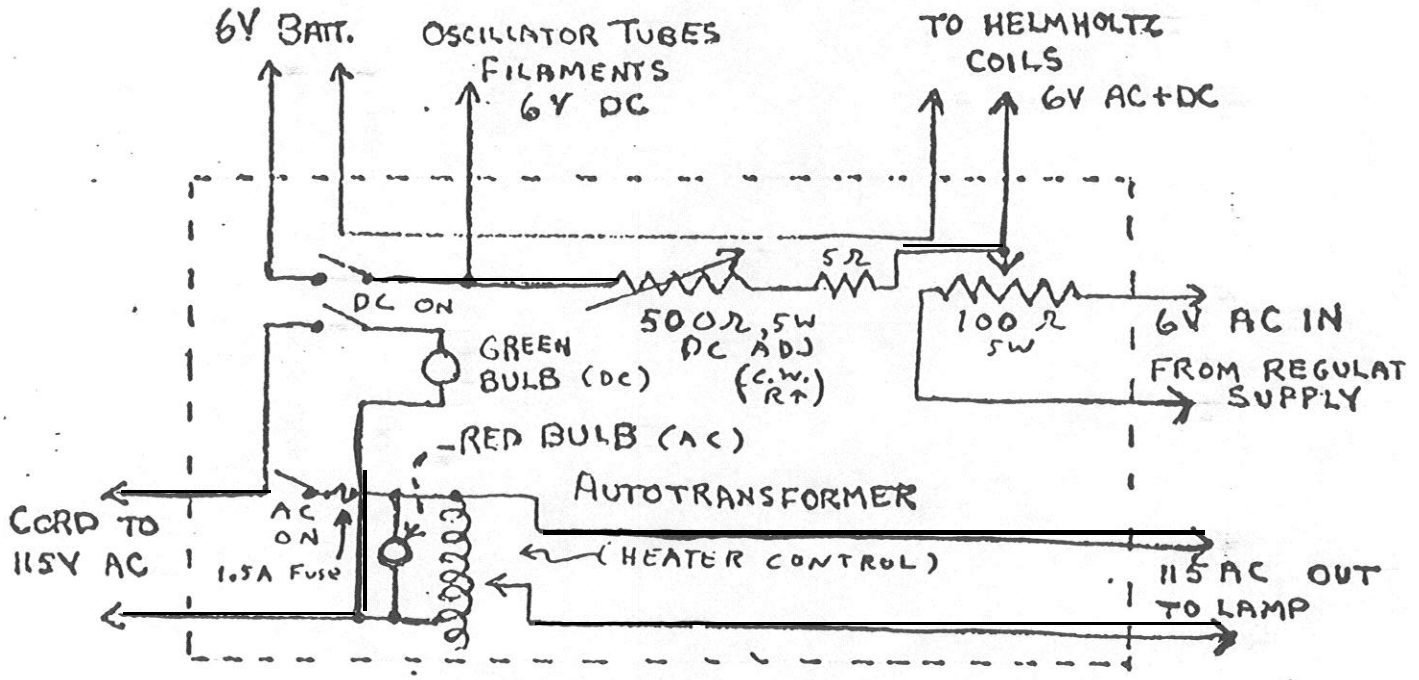
Fig. 2





MODULATION AND HEATER CIRCUIT BOX (CONTROL UNIT)





MODULATION AND HEATER CIRCUIT BOX
SCHEME

Fig. 3B

light carries one quanta of angular momentum. In order to conserve angular momentum in the Rb bulb, the only allowable transitions for a Rb atom which absorbs a photo" of circularly polarized light are those for which $\Delta M_J = \pm 1$. The sign of ΔM_J is dependent on the nature of the circular polarization--right-handed or left handed polarization. For the case that $\Delta M_J = +1$ the possible absorption and. re-emission transitions are shown in Fig. 1 by the arrows (implicit is the fact that $\Delta M_J = 0$ or ± 1 for ordinary--electric dipole--optical transitions). Since the D_2 light is stopped by the D_1-pass filter, none of the $^2P_{3/2}$ states are excited. Because of the selection rule, atoms in the A state are excited , to the C state. Some of the excited atom6 decay back to the A state and the remainder decay to the B state. Light from these transitions is emitted in all directions from the Rb bulb. The atoms left in the A state are again excited to the C state. Jf there were no way for atoms to leave the B state, all the atoms would eventually be pumped into the B state.

However, when a Rb atom strikes the glass wall of the resonance bulb, the probability that an atom. In state B will "relax" into state A is very high. I" order to reduce the disorienting or relaxing effects of wall collisions, the cell is filled with a few mm of Hg of some buffer gas, usually one of the inert gases He, A, or Kr. The relaxation time caused by wall collisions is the" much longer than the pumping time of the light and a large polarization or population difference results.

A transition from B to A is stimulated when an rf field at a frequency corresponding to the energy difference between A and B is applied to the absorption cell. Such a field equalizes the populations. The optical absorption immediately increases since atoms in A can absorb the light. The resonance can then he detected by monitoring the transmitted light as in Fig. 5, 6.

The states A and B in this discussion play a role analagous to the states - and + respectively in the discussion reproduced above from Experiment 6 (the atomic, or molecular, states with the electron magnetic moment parallel and antiparallel to the magnetic field), and the numbers of atoms "A andnB in states A and B replace the numbers n_- and n+ respectively in that discussion. In Experiment 6 the numbers of atoms in the two states are determined by the conditions of thermal equilibrium, and more atoms are in the state of lower enerey ($n_- > n_+$); the magnetic field is large to maximize the energy difference between the states and therefore the fractional difference in the occupation numbers, so that the ebsorotion of microwaves is maximized, but even so the fractional difference in occupation numbers is quite small. In the present experiment the numbers of atoms in the two states in the absorption cell are not thermal but are determined by the absorotion of the pumping optical radiation, and most of the atoms are in the state of higher energy; the fractional difference in occupation numbers is near one, therefore, and the emission stimulated by the rf field, whose role is analagous to that of the microwaves in Experiment 6, is therefore large. This accounts for an improvement of a factor of about 100 in sensitivity of optical pumping over microwave absorotion, which was mentioned in (b) and (c) of the material from Experiment 6. Since a large energy difference is not required here, a small magnetic field can be used, in which case rf radiation can be used in place of microwave radiation.

(b) Experimental Details

The natural Rb lamp is placed in an oven, and it is excited by an rf electrodeless discharge at about 30 Mc/sec.[6] The resonance cell contains natural Rb (72.15% Rb^{85} and 27.85% Rb^{87}) and a carrier gas. The transmitted light is detected by a silicon detector and common base amplifier (Fig.4). A magnetic field of about one gauss can be imposed in the Rb bulb (absortion cell) along the direction of the light beam by a set of Helmholtz coils which producew a relatively uniform field across the cell. The magnetic field is modulated by applying an AC voltage component to the Helmholtz coils. This provides for easier-detection of the resonance peaks as shown in Fig 6 (Another set of coiles perpendicular to

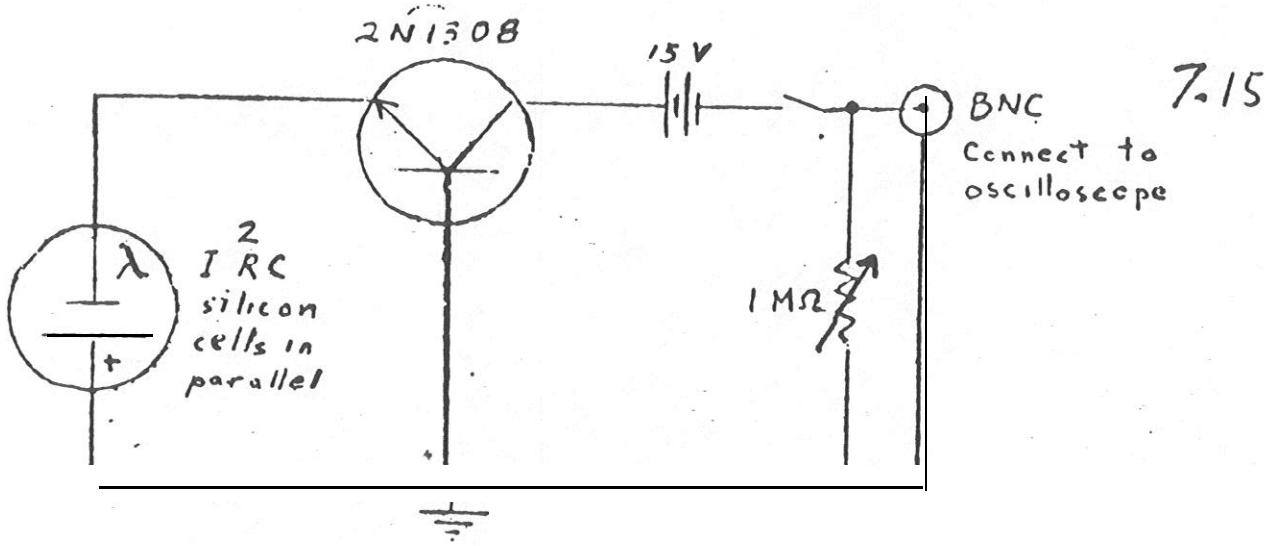
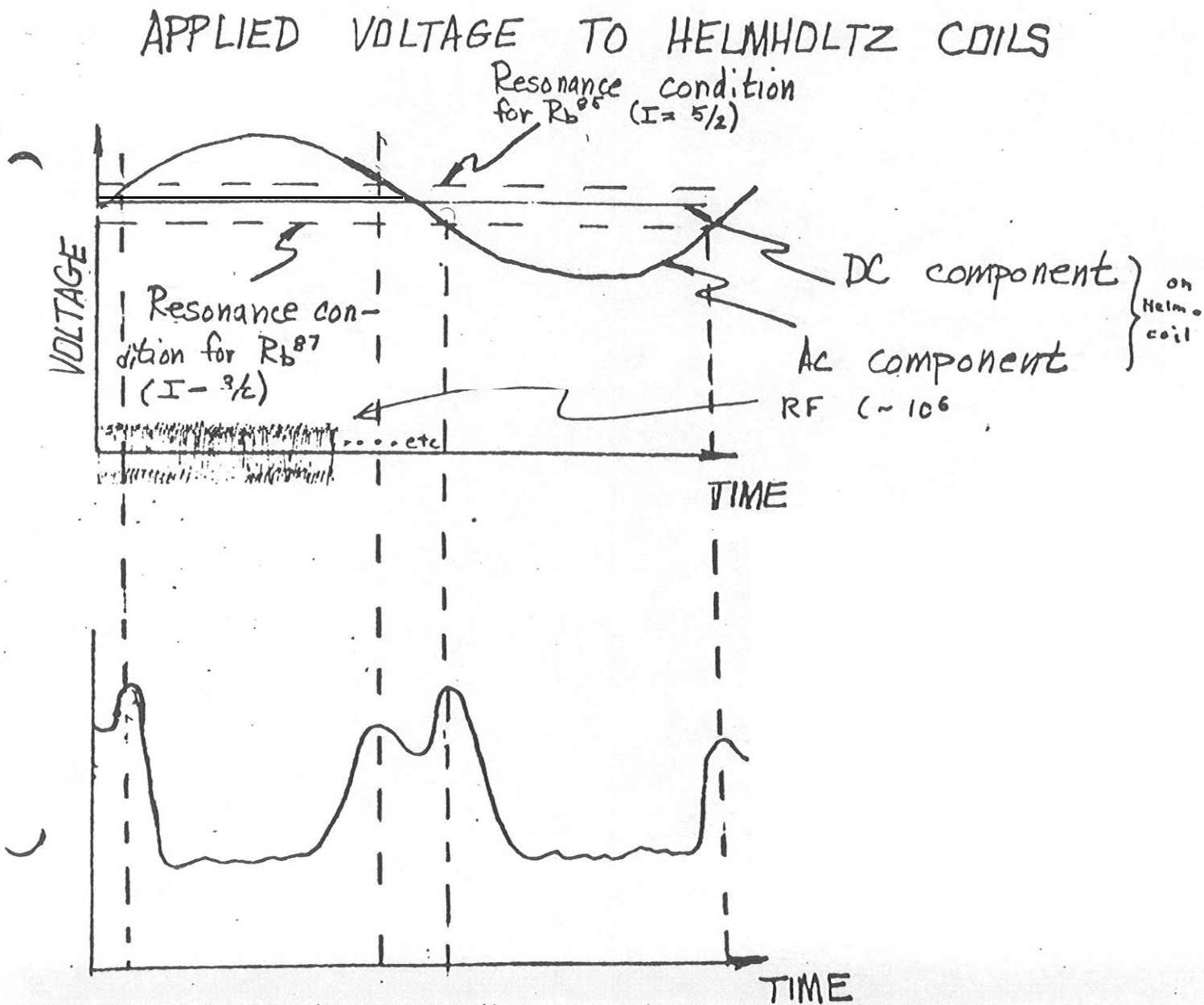


Figure 4

SIMPLE

Optical Pumping **Detector** and^Amplifier



APPLIED VOLTAGE TO HELMHOLTZ COILS

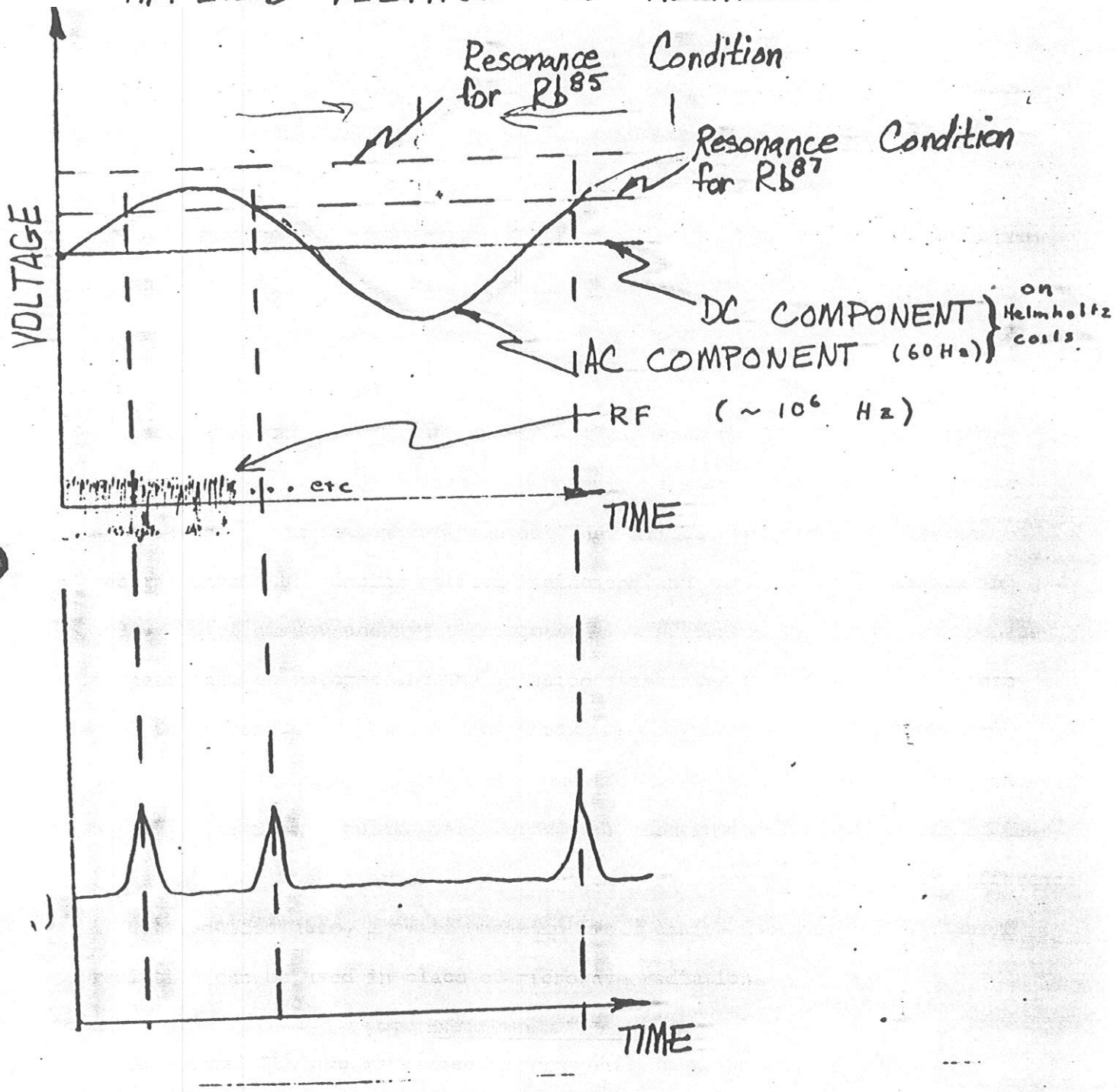


Fig. 6

the beam allows an rf field to be applied to the cell.

Suggested Procedure (Also listen to tape recording, which is more recent)

1. connect the circuit as in Fig. 3.
2. Turn on all of the equipment and allow it to warm up. Use a Tektronix scope to detect the absorption resonances as the Heathkit scopes are often insensitive to the weak signals. The Heathkit power supply should be turned to standby. Both the AC and the DC switches on the control unit should be turned on. The AC switch connects power to the lamp heater. The DC switch connects power to the filaments of the oscillator tubes. enclosed in the lamp box and to the Helmholtr coils.
3. The Rb lamps should be hot for start-up. The rheostat which controls the oven temperature should be set for, a maximum during an initial five minute warm-up period. After this warm-up period the setting on the rheostat can be reduced to an operating level--usually between 30 and 60. The operating level is determined by the color of the lamp as explained below.
4. After the initial warm up period, start the lamp by turning on the B+ voltage of the Heathkit power supply. Increase the voltage from zero to a voltage at which the oscillator tubes in the lamp box are drawing 25 ma of current (perhaps slightly more at start). Do not exceed either 25 ma or 175 volts (after start). If the tubes draw no current at 175 volts, recheck your connections.
5. Throughout the experiment you should endeavor to maintain the Rb light at a bright deep pinkish-purple color., The color is achieved by adjusting the temperature of the lamps. When the lamp is either too hot or too cold and dim the color of the lamp is light pink^ 'The light color is the result of Kr discharge which conceals the Rb discharge. When this occurs, it is highly unlikely that resonance signals will be detected, so it is important to achieve and maintain the deep pinkish-purple color.
6. Now you-should build up the optical bench. Lens 1 should focus the Rb lamp image onto the absorption cell. Replacing the cell with a piece of paper will aid in achieving the focus. Lens 2 should focus this image onto the Si detector. Then the filter and polarizer should be put into place YOU should he careful to make sure that all of the light reaching the detector passes through the filter, the polarizer and the absorption cell. Furthermore, since the resonance signal will he very weak, check to insure that no stray light reaches the detector.

7. It is necessary to make a preliminary adjustment of the applied magnetic field. Measure-the voltage across the resistor which is in series with the Helmholtz coils with the Tektronix oscilliscope.* Set the input of the scope on DC, and trigger the scope on line. Using (1) calculate the magnetic field required to see a resonance at an rf frequency between 0.5 and 1 mc/sec for Rb^{87} (I = 3/2) and Rb^{85} (I = 5/2). Set the DC component of the voltage to the desired value. (See the e/m write-up for the Helmholtz coil formula.* The number of terms is indicated on the coil.) Adjust the AC component so that the peak to peak voltage is about one fourth of the applied DC voltage.
8. Apply the rf signal from. the sine-wave generator to the absorption cell. For early success, set the output of the generator at a maximum.
9. To detect an absorption resonance, monitor the output of the detector while you sweep the rf spectrum with the signal generator. The output of the amplifier is monitored with the oscilloscope. The input of the scope is set for AC and a sensitivity in the millivolt region. The scope is triggered on line. A resonance appears as a peak in the scopes pattern. A true resonance disappears if the circular polarizer is reversed or if the DC component of the Helmholtz coil current is changed appreciably. ESR has proven to be an accurate instrument for the measurement of weak magnetic fields.' Using (1) with the observed resonance frequencies you can determine the magnitude of the applied field on the atoms. By reversing the magnetic field of the Helmholtz coils you can observe the efect of the earth's magnetic field on the Rb atoms in the absorption cell.

* $B = \frac{\mu_0 N I}{a} \frac{8}{5^{3/2}} = .9 \times 10^{-6} \frac{NI}{a}$ w/m²

for pair of coils of N turns each, radius a in meters, and current I in amperes.

* If you do this, check the value of the resistance by measurement. Alternatively use an ammeter if one is available when you need it.

(c) ESR Optical Pumping--Lab Write Up

1. Draw the energy diagram for Rb with I = 0, include $^2S_{1/2}$, $^2P_{1/2}$, $^2P_{3/2}$ lines. Explain what happens in the pumping process.
2. Measure the resonant frequencies for the two Rb isotopes, at one setting of the magnetic field. Reverse the magnetic field and measure the resonant frequencies again sing the same DC voltage in the Helmholtz coils. From your data calculate the magnitude of the magnetic fields. What is the error in your measurements? What contributed to the error? What is the size of the component of the earth's magnetic field along the direction of the optical bench?
3. When the atoms have been pumped into the B state-- ^{2}&,,+ 1/2 state, why don't they absorb photons from the Rb source and go to the $^2P_{1/2}$,- 1/2 state?

(d) Equipment List

Resonance cell
7947 R-pass filter
Helmholtz coils and rf coils
Sine-wave generator
191 Tektronix signal generator and 502 Tektronix oscilloscope to be shared by all optical pumping set-ups. Manual for the 191 is in 274 LeConte.
Circular polarizer
Two 17.5 cm lenses and holders
Optical bench
Rb lamp and cables
Modulation and heater circuit
Silicon detector and amplifier
Black cloth
Regulated power supply
Oscilloscope
6 V storage battery
Two 3' BNC cables

(e) References

1. A. L. Bloom, Scientific American 203, 72 (October 1960).

2. R. L. de Zafra, American Journal of Physics 28, 646 (1960).

3. T. R. Carver, Science 141, 599 (1963).

4. A. L. Bloom, Applied Optics 1, 61 (1962).

5. G. V. Skrotskii and T. G. Izyumova, Soviet Physics Uspekhi 4, 177 (1961).

6. W. E. Bell, A. L. Bloom, and J. Lynch, Review of Scientific Instruments 32, 688 (1961).

7. E. H. Wichmann, Perkeley Physics Course, Vol. IV - Quantum Physics, Chapter 3.

8. R. Benumof, American Journal of Physics 33, 151 (1965).

9. A. M. Portis, Laboratory Physics (McGraw-Hill, 1966), Parts C and D, Experiment D-5.

10. P. A. Tipler, Foundations of Modern Physics, pp. 296-301 and p. 346. From Experiment #6:

11. G. E. Pake, Scientific American 199, 58 (August, 1958)

12. C. P. Slichter, Principles of Magnetic Resonance, portions of Chaps. 1 & 2

13. R. E. Rorberg, "Resource Letter NMR-EPR-1 on Nuclear Magnetic Resonance", Am. J. Phys. 33, 71 (1965)

14. A. M. Portis, Laboratory Physics (McGraw-Hill, 1965), Part B pp. B9-79, Parts C and D, Experiment D-4.

(f) Appendix: rf resonance condition

In a low magnetic field the hyperfine interaction couples the electron and nuclear angular momenta: $\vec{F} = \vec{J} + \vec{I}$.
total elect. nuc.

The energy of an atom with magnetic dipole moment $\vec{\mu}$ is

$$E = \langle -\vec{\mu} \cdot \vec{H} \rangle \ .$$

But $\vec{\mu} = \vec{\mu}_J + \vec{\mu}_I$ or $E = -\langle \vec{\mu}_J \cdot \vec{H} \rangle - \langle \vec{\mu}_I \cdot \vec{H} \rangle$. But one can always write $\langle \vec{\mu} \cdot \vec{H} \rangle$ as $\mu_o g_F \langle \vec{F} \cdot \vec{H} \rangle$ where

$$\mu_o = \frac{e\hbar}{2mc} = 9.2 \times 10^{-21} \text{ ergs/gauss}$$

is the Bohr magneton and g_F is defined such that the above substitution is valid. Then

$$E = -g_F \mu_o \langle \vec{F} \cdot \vec{H} \rangle = -g_J \mu_o \langle \vec{J} \cdot \vec{H} \rangle - g_I \mu_o \langle \vec{I} \cdot \vec{H} \rangle$$

a. or $\langle (g_F \vec{F} - g_J \vec{J} - g_I \vec{I}) \cdot \vec{H} \rangle = 0$ for any $\vec{H}$. Then

$$\langle (g_F \vec{F} - g_J \vec{J} - g_I \vec{I}) \cdot \vec{F} \rangle = 0 \text{ also.}$$

$$g_F F(F+1) - g_J \langle \vec{J} \cdot \vec{F} \rangle - g_I \langle \vec{I} \cdot \vec{F} \rangle = 0$$

$$\vec{F} = \vec{I} + \vec{J}, \quad \vec{F} \cdot \vec{F} = \vec{I} \cdot \vec{I} + \vec{J} \cdot \vec{J} + 2\vec{I} \cdot \vec{J}$$

$$\xrightarrow[\text{mechanics}]{\text{quantum}} F(F+1) = I(I+1) + J(J+1) + 2\langle \vec{I} \cdot \vec{J} \rangle$$

$$\Rightarrow \langle \vec{I} \cdot \vec{J} \rangle = [F(F+1) - I(I+1) - J(J+1)]/2$$

Similarly one can find $\langle \vec{J} \cdot \vec{F} \rangle$ and $\langle \vec{I} \cdot \vec{F} \rangle$. Then

$$g_F = \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)} g_J + \frac{F(F+1) + I(I+1) - J(J+1)}{2F(F+1)} g_I$$

Since $g_I/g_J \approx m_{electron}/m_{proton}$ the last term can be neglected here since the first term does not vanish. In the same manner, one has $\vec{J} = \vec{L} + \vec{S}$ (total = orb. + spin elect.) and

$$g_J = \frac{J(J+1) + L(L+1) - S(S+1)}{2J(J+1)} g_L + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)} g_S$$

$\longrightarrow g_S$ for $L = 0$ & $J = S = 1/2$

For $J = 1/2$, $F = I \pm 1/2$:

$$g_F \approx \frac{(I \pm 1/2)(I \pm 1/2 + 1) + 3/4 - I(I+1)}{2(I \pm 1/2)(I \pm 1/2 + 1)} g_S = \frac{\pm g_S}{2I+1}$$

$$E = -g_F \mu_o \langle \vec{F} \cdot \vec{H} \rangle = -g_F \mu_o H M_F$$

$$\Delta E = |E_{M_F+1} - E_{M_F}| = g_F \mu_o H$$

$$\frac{\nu}{H} = \frac{\Delta E}{hH} = \frac{g_F \mu_o}{h} = \left(\frac{\pm g_S}{2I+1}\right) \frac{9.2 \times 10^{-21} \text{ erg/gauss}}{6.625 \times 10^{-27} \text{ erg-sec}}$$

Since $g_S \approx 2$,

$$\frac{\nu}{H} = \frac{2.8}{2I+1} \text{ Hc/sec gauss}$$